

Chaniel Abou-Ezzi

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Education

University of Massachusetts Amherst

September 2022 – May 2025

Bachelor of Science in Computer Science, Summa Cum Laude

Amherst, MA

- GPA: 3.99/4.0
- Relevant Courses: Linear Algebra, Probability, Multivariable Calculus, Discrete Math, Systems Principles, Intro to Algorithms, Computer Graphics, Intro to Simulation, Quantum Information Science, Machine Learning, Software Engineering, Algorithms for Data Science
- Activity: Peer Mentor, Arab Cultural Association, Chess Club

Experience

iOS Software Engineer

August 2025 – Present

Apple

Austin, TX

Worked on the frontend of the EasyPay app in Retail Engineering, the POS software for Apple Retail. Work included creating LLM-based tooling for frontend development.

Software Engineering Intern

May 2024 – August 2024

Apple

Sunnyvale, CA

Summer internship on the Cellular Platform Software team in Wireless Technologies Engineering. Pioneered a new internal tool for summarizing, comparing, and answering questions related to hardware specifications.

- Built off the Retrieval-Augmented Generation (RAG) architecture
- Used BM42 hybrid search, cross-encoder reranking to improve retrieved information relevance (+40% accuracy)
- Improved thematic responses with Recursive Abstractive Processing for Tree-Organized Retrieval (+20% accuracy)
- Used TruLens framework for app monitoring and evaluation

iOS Software Development Intern

June 2023 – September 2023

Wolters-Kluwer

Waltham, MA

Remote summer internship working as a developer for the Lexicomp iOS mobile app. Work consisted mainly of enhancing legacy code, working on new UI features, and maintenance of local data handling.

Research

University of Massachusetts Amherst

December 2024 – February 2025

Undergraduate Research Volunteer

Amherst, MA

Studied and implemented algorithms for stochastic and adversarial multi-armed bandits.

- Studied UCB, EXP-3, & FTRL/Tsallis-regularized online mirror descent algorithms
- Implemented simulation experiments in Python to compare algorithm behavior across **stochastic vs. adversarial** reward settings; generated plots for visualization

Projects

Find demos and more at <https://cabouezzi.duckdns.org:8443>

GPT-2 from Scratch | Python, PyTorch

January 2025

Implemented GPT-2 from scratch, including tokenizer, model architecture, training loop, and inference meant for running on my M1 Pro chip.

- Implemented decoder-only GPT-style **Transformer in PyTorch** (6 layers, 6 heads, 384-dim embeddings, 256 context) with causal masked self-attention, MLP blocks, layer norms, and positional embeddings
- Built a character-level tokenizer and data pipeline (*stoi/itos*), including train/val split and randomized batch sampling for next-token prediction
- Wrote full training loop with **AdamW**, periodic train/val loss evaluation, and checkpoint saving for reproducible runs and resuming
- Implemented **inference** that loads checkpoints and **streams** sampled tokens in real time to *stdout* and an output log file.
- Optimized for local training on **Apple Silicon** by using the MPS backend when available.

PhotoManagement | Python, ChromaDB, Embeddings

October 2024 – December 2024

Group project developing a local photo library API and GUI with perceptual duplicate detection, semantic search, and image-to-audio descriptions.

- Local photo ingestion pipeline for JPG/PNG directories, maintaining metadata and embeddings for fast retrieval
- Used ChromaDB (SQLite-backed) to store **CLIP embeddings** for semantic similarity search and text-to-image retrieval
- Implemented “similar photo search” using **perceptual hashing** and embedding similarity thresholds
- Integrated a **multimodal vision LLM** via Ollama for image captions and connected captions to **text-to-speech** for audio playback
- Shipped as a pip-installable package with a cross-platform GUI, a tutorial notebook, and generated HTML documentation

ML Playground | Python, PyTorch, Jax, Scikit-learn

September 2024 – December 2024

Projects from CS589, the graduate machine learning course taught at UMass.

- Implemented **Bayesian linear & logistic regression** using SciPy to derive & sample posterior distributions
- Used JAX to implement **backpropagation, gradient descent**, and train and run neural networks
- Implemented **K-Means & PCA pipelines** for lossy image compression and quantitative reconstruction analysis
- Developed decision tree, and **K-NN classifiers** with custom K-fold cross-validation on image datasets
- Implemented a **Multi-Layer Perception (MLP)** with configurable layers and activations, optimized via gradient descent

Lira Playground (WASM) | C#, .NET WASM, React

August 2023

Browser-based IDE for a custom programming language, running entirely client-side via WebAssembly.

- Implemented **lexer/parser and AST interpreter** in C#, exposing a clean execution interface for the UI
- Compiled the runtime to .NET WebAssembly and built a React-based **editor and console** for live demonstration (see portfolio for demo)
- Added **error reporting** and structured output to support interactive demos and debugging

Interactive Weather | C#, Unity, ShaderLab

February 2022

3D globe visualization of global wind and temperature patterns using NOAA atmospheric data.

- Processed and aggregated **wind-vector and temperature datasets** into renderable layers on a **3D globe**
- Wrote **shaders/animations** for wind flow and temperature coloration; optimized for interactive exploration
- Built interaction controls (rotate/zoom) for an educational visualization experience

Smart Andover App | Swift, SwiftUI, Firebase

September 2021

iOS app for a sustainability club: submissions, moderation, points leaderboard, and reward redemption.

- Built **authentication** and **Firestore-backed data model** for users, submissions, and point transactions
- Implemented **photo submission with compression** and an admin review flow (accept/reject + feedback)
- Added leaderboards and redemption tracking to gamify sustainable actions

Chess AI | Swift

July 2021 – August 2021

Implemented a chess engine from scratch and corresponding macOS GUI. A fixed-depth version of the engine used to compete at lichess.org/@/cabouezzi, made using the Lichess Bot API and running on Ubuntu in Oracle Cloud.

- Hybrid representation of **Bitboards and piece lists**, using **bitwise operations** to calculate legal moves
- **Negamax** search algorithm with **Alpha-Beta pruning** technique, with **hash table** for chess openings

Boids | Swift

January 2020

iOS/macOS flocking simulation package implementing separation, alignment, and cohesion behaviors. Used in BeeKeeper (see below).

- Implemented a **real-time** boids update loop with steering forces, boundary handling, and tunable parameters
- Built a renderer and interaction controls to explore emergent behavior

BeeKeeper | Swift, SceneKit/SpriteKit, Blender

January 2018 – June 2025

iOS beekeeping simulator game featuring an agent-based bee simulation and evolution/upgrade mechanics.

- Designed building mechanics and relationship patterns, for example bees assigned to flowers or hives, transferring resources from hives to storage, etc.
- Built a **simulation** game ticker for production and **event probabilities**; persisted player state and progression

- Implemented boids-based bee movement and validated an evolution graph for upgrade paths
- Created **3D assets and animations** individually in **Blender**, created some 2D assets and the rest generated by AI later in 2025
- Shipped the app publicly in 2025 (**App Store release**), just before I joined Apple

Matching Coco | *Swift, UIKit*

June 2017

[Scene: I am in 7th grade] *First iOS project: card-matching game with original photos of my cousin's dog, Coco.*

- Implemented game state, scoring, and basic UI animations in UIKit as a first end-to-end iOS build

Technical Skills

Programming Languages: Swift, C, Python, C++, C#, JavaScript/TypeScript, Java

AI/ML/DS: Pandas, Numpy, Matplotlib, Vega-Altair, TruLens, Langchain, PyTorch, Jax, Scikit-learn, Ollama

iOS/macOS Development: UIKit/AppKit, SwiftUI, CloudKit, MapKit, SceneKit/SpriteKit, CocoaPods, SwiftPM

Database Services: Firestore, MongoDB, Qdrant, Chroma

Art: Inkscape, Linearity Curve, Blender

Foreign Languages: Arabic, Spanish